

Table 1: **Out-of-sample multi-view clustering.** Inductive AMI on held-out points, mean $\pm$ std over the 19 dataset means (10 seeds each, 190 cells per method). Inductive AMI clusters the *training* embedding with k -means and applies that clustering to the extended test embedding, so it fails unless the extension lands unseen points in the same coordinate frame. All diffusion methods share one kernel, one truncated-SVD embedding at $k = |C|$ components, and one Nyström extension obtained by writing each operator as $\mathbf{W} = \sum_v \mathbf{P}_v \mathbf{S}_v$ and substituting the test-to-train transition for the leftmost factor. * inside the MDT operator space (Sec. 3.5); a win over these concerns trajectory choice, not the framework. $\S$ inductive by construction, no Nyström step. PRR is Eq. 1 against MDT-RAND as a ratio of means. $W-L$ counts datasets on which MDT-DIRECT beats / loses to that row; p_H is a Holm-corrected exact sign test over the 16-comparison family. † ours, not a variant of the paper. ‡ per-cell maximum over the six MDT variants; it uses test labels, so it is an upper bound rather than a method, and its sign test is degenerate by construction. ¶ concatenation projected to k dimensions, so that the control has the same representational budget as the diffusion arms.

Method	Ind. AMI	Test AMI	PRR	W-L	p_H	Time (s)
<i>MDT variants (paper Tab. 2)</i>						
MDT-DIRECT*	0.417$\pm$0.329	0.436	1.25	–	–	21.6
MDT-CVX-RAND*	0.380 $\pm$ 0.308	0.403	1.14	14–5	0.445	0.1
MDT-BSC*	0.370 $\pm$ 0.313	0.386	1.11	18–1	0.001	18.1
MDT-CST*	0.348 $\pm$ 0.269	0.365	1.04	17–2	0.010	1.5
MDT-SIL ‡ *	0.338 $\pm$ 0.287	0.347	1.01	18–1	0.001	11.0
MDT-RAND*	0.333 $\pm$ 0.269	0.337	1.00	16–3	0.049	0.1
MDT best-of-6 ‡ *	0.455 $\pm$ 0.310	0.477	1.37	n/a	–	–
<i>Operator-based multi-view diffusion (paper Tab. 1)</i>						
AD*	0.320 $\pm$ 0.298	0.321	0.96	16–3	0.049	0.1
P-AD*	0.322 $\pm$ 0.294	0.323	0.97	17–2	0.010	1.4
ID*	0.309 $\pm$ 0.290	0.322	0.93	17–2	0.010	0.8
MVD	0.320 $\pm$ 0.298	0.330	0.96	18–1	0.001	1.3
CR-DIFF	0.317 $\pm$ 0.274	0.286	0.95	16–3	0.049	0.8
COM-DIFF (4 sets)	0.076 $\pm$ 0.068	0.093	0.76	4–0	0.625	0.1
<i>Non-diffusion, inductive by construction</i>						
SpecRaGE §	0.417 $\pm$ 0.276	0.436	1.25	10–9	1.000	47.4
DGCCA §	0.381 $\pm$ 0.287	0.434	1.15	11–8	1.000	2.9
GCCA §	0.323 $\pm$ 0.290	0.384	0.97	15–4	0.154	0.2
<i>Controls</i>						
Uniform mean*	0.413 $\pm$ 0.327	0.441	1.24	9–10	1.000	0.1
Concat. features §	0.355 $\pm$ 0.331	0.350	1.07	13–6	0.668	0.0
Concat. features (k -dim) ¶ §	0.341 $\pm$ 0.278	0.381	1.02	14–5	0.445	0.0

MDT out of sample: numbers sheet

$$\text{PRR}(\text{method}) = \frac{\text{AMI}(\text{method})}{\text{AMI}(\text{MDT-RAND})} \quad (1)$$

Table 2: **Per-dataset inductive AMI**, mean over 10 seeds. Best per row in bold. Three datasets (Movies, NUS-WIDE, ProteinFold) sit near the AMI floor for every method; on five others (3Sources, BBCSport, Prokaryotic, Reuters-1200, WebKB) the MDT variants fall below 0.11 while another method clears 0.15, so those are method failures rather than uninformative data.

Dataset	MDT-DIRECT	MDT-CVX-RAND	Uniform mean	SpecRaGE	AD	ID	MVD	CR-DIFF	GCCA
100Leaves	0.817	0.628	0.815	0.602	0.549	0.553	0.579	0.487	0.177
3Sources	0.066	0.048	0.043	0.202	0.021	0.017	0.040	0.015	-0.003
ALOI	0.574	0.594	0.626	0.597	0.473	0.474	0.493	0.476	0.193
BBCSport	0.106	0.107	0.112	0.157	0.075	0.090	0.071	0.083	0.022
Caltech101-7	0.515	0.501	0.525	0.564	0.352	0.253	0.330	0.432	0.483
Handwritten	0.883	0.869	0.882	0.880	0.830	0.816	0.833	0.796	0.762
MNIST-10k	0.739	0.731	0.744	0.653	0.702	0.651	0.680	0.633	0.669
MNIST-4	0.782	0.752	0.779	0.728	0.731	0.708	0.718	0.781	0.724
MSRC-v5	0.630	0.585	0.663	0.576	0.250	0.159	0.255	0.509	0.642
Movies	0.034	0.027	0.015	0.092	0.037	0.025	0.020	-0.001	0.005
NUS-WIDE	0.090	0.092	0.108	0.099	0.063	0.071	0.070	0.058	0.108
OutdoorScene	0.563	0.518	0.568	0.549	0.477	0.466	0.485	0.363	0.487
Prokaryotic	0.035	0.042	0.028	0.113	0.022	0.020	0.018	0.037	0.240
ProteinFold	0.008	0.039	0.055	0.129	0.050	0.086	0.108	0.042	0.032
Reuters-1200	0.073	0.064	0.056	0.240	0.053	0.073	0.051	0.043	0.011
UCI	0.871	0.796	0.835	0.859	0.841	0.837	0.835	0.628	0.763
WebKB	0.099	0.091	0.129	0.045	0.138	0.150	0.068	0.126	0.002
Wikipedia	0.518	0.217	0.346	0.509	0.039	0.043	0.039	0.170	0.395
Yale	0.521	0.521	0.525	0.331	0.376	0.384	0.394	0.345	0.420

Table 3: **The paper’s constructed datasets**, inductive AMI, mean over 10 seeds. Added so the evaluation spans 8 of the 9 clustering datasets of paper Tab. 3. View construction follows the reference loaders exactly; PCA and min-max scaling are refitted on the training rows only, because upstream fits them on all samples before any split. L-Isolet is excluded: its three views are three *different kernels* on one feature matrix, and this benchmark controls the kernel.

Method	K-MvMNIST	L-MvMNIST	Olivetti	Mean
Uniform mean	0.382	0.353	0.370	0.368
MDT-DIRECT	0.380	0.346	0.348	0.358
Concat. features	0.375	0.371	0.306	0.351
MDT-CVX-RAND	0.369	0.332	0.340	0.347
GCCA	0.344	0.346	0.313	0.335
P-AD	0.365	0.330	0.294	0.329
MDT-SIL	0.355	0.292	0.321	0.323
ID	0.340	0.303	0.298	0.314
Concat. features (k -dim)	0.307	0.323	0.305	0.311
AD	0.348	0.273	0.304	0.308
MDT-BSC	0.347	0.286	0.286	0.306
MDT-RAND	0.336	0.274	0.307	0.305
DGCCA	0.287	0.298	0.330	0.305
MDT-CST	0.322	0.253	0.311	0.295
MVD	0.269	0.257	0.272	0.266
CR-DIFF	0.322	0.242	0.198	0.254

Table 4: **Train-fraction sensitivity.** Inductive AMI over the 19 datasets, seeds 0–4 in all three columns. The out-of-sample ordering is stable: MDT-DIRECT beats every published operator-based competitor at all three fractions, and its tie with the uniform-mean trajectory is unchanged (+0.006, +0.003, −0.004).

Method	0.5	0.7	0.9
MDT-DIRECT	0.418	0.415	0.398
Uniform mean	0.412	0.412	0.402
DGCCA	0.384	0.384	0.382
MDT-CVX-RAND	0.387	0.382	0.369
MDT-Bsc	0.388	0.373	0.374
Concat. features	0.356	0.360	0.355
MDT-RAND	0.330	0.334	0.318
GCCA	0.331	0.329	0.323
P-AD	0.330	0.321	0.317
MVD	0.325	0.320	0.333
AD	0.324	0.319	0.319
CR-DIFF	0.316	0.319	0.312
ID	0.321	0.308	0.304

Table 5: **Kernel ablation.** Inductive AMI, 19 datasets $\times$ 5 seeds, paired cell for cell. *Scale-inv.* is $\exp(-d^2/2\sigma^2)$ with σ a median pairwise distance on standardised views: dimensionless, hence invariant to rescaling a view. *Reference* reproduces `benchmarks/utilities.get_kernel_matrix` of the MDT reference repository: $\exp(-d^2/bw)$ with $bw = \max_i \min_{j \neq i} d_{ij}$ on raw views. That divides a squared distance by a distance, so it is not scale invariant and on views with large absolute scale every off-diagonal weight underflows and the row-normalised operator becomes the identity (mean diagonal 1.000 on MSRC-v5 view 1, 0.997 on Handwritten view 1, 0.917 on Yale view 1); estimating the bandwidth on training rows only is not the cause, train and full-data bandwidths agreeing to 1–3%. Every diffusion arm degrades and the ordering does not survive (Spearman $\rho = +0.525$), so the main result is kernel-conditional.

Method	Scale-inv.	Reference	Δ	Ref. wins	p
MDT-DIRECT	0.415	0.356	-0.059	6/19	0.1671
MDT-CVX-RAND	0.382	0.331	-0.051	5/19	0.0636
MDT-Bsc	0.373	0.273	-0.100	6/19	0.1671
MDT-CST	0.348	0.242	-0.106	3/19	0.0044
MDT-SIL	0.337	0.147	-0.190	5/19	0.0636
MDT-RAND	0.334	0.274	-0.060	4/19	0.0192
AD	0.319	0.281	-0.038	5/19	0.0636
P-AD	0.321	0.270	-0.052	4/19	0.0192
ID	0.308	0.265	-0.044	7/19	0.3593
MVD	0.320	0.282	-0.038	5/19	0.0636
CR-DIFF	0.319	0.323	+0.005	7/19	0.3593
DGCCA	0.384	0.399	+0.014	14/19	0.0636
GCCA	0.329	0.328	-0.001	12/19	0.2379
Uniform mean	0.412	0.355	-0.057	5/19	0.0636
Concat. features	0.360	0.360	+0.000	0/19	–

Table 6: **Noise axis** (analogue of paper Fig. 7), $n_{\text{train}} = 2000$, 5 seeds, inductive AMI. s is the noise parameter. In K-MvMNIST *only the second view degrades*, so it isolates whether a method can down-weight a deteriorating view; in L-MvMNIST both views degrade. *Decay* is AMI at $s=0.1$ minus AMI at $s=0.9$; smaller is more robust. The searched MDT variants are omitted: MDT-DIRECT requires 100 objective evaluations per cell, each a full $n \times n$ SVD. MDT-Cvx-RAND minus the uniform mean on K-MvMNIST is $-0.001, +0.005, +0.005, -0.009, -0.014$ across the five levels, with every 95% interval covering zero: the view-weighting mechanism shows no advantage where it should be strongest.

<i>K-MvMNIST – only view 2 degrades</i>						
Method	$s=0.1$	$s=0.3$	$s=0.5$	$s=0.7$	$s=0.9$	Decay
Uniform mean	0.443	0.432	0.435	0.446	0.428	+0.015
MDT-Cvx-RAND	0.442	0.436	0.440	0.437	0.414	+0.028
ID	0.425	0.422	0.421	0.422	0.433	-0.008
P-AD	0.425	0.423	0.410	0.445	0.427	-0.003
MDT-RAND	0.420	0.413	0.406	0.362	0.289	+0.130
CR-DIFF	0.407	0.398	0.381	0.335	0.308	+0.099
Concat. features	0.395	0.429	0.445	0.477	0.445	-0.050
AD	0.394	0.400	0.408	0.396	0.405	-0.011
GCCA	0.369	0.360	0.354	0.361	0.357	+0.013
MVD	0.365	0.351	0.371	0.346	0.356	+0.010
DGCCA	0.277	0.332	0.352	0.358	0.368	-0.091
Concat. features (k -dim)	0.262	0.286	0.337	0.385	0.403	-0.141
<i>L-MvMNIST – both views degrade</i>						
ID	0.442	0.418	0.368	0.272	0.093	+0.349
P-AD	0.429	0.402	0.356	0.283	0.090	+0.339
MDT-Cvx-RAND	0.427	0.407	0.383	0.347	0.107	+0.320
MDT-RAND	0.424	0.373	0.356	0.249	0.068	+0.356
Uniform mean	0.424	0.394	0.379	0.358	0.113	+0.311
AD	0.417	0.381	0.325	0.239	0.077	+0.339
CR-DIFF	0.413	0.354	0.299	0.223	0.083	+0.330
Concat. features	0.402	0.409	0.458	0.431	0.184	+0.218
MVD	0.376	0.362	0.299	0.248	0.076	+0.300
GCCA	0.375	0.362	0.350	0.362	0.279	+0.097
Concat. features (k -dim)	0.243	0.285	0.380	0.389	0.315	-0.073
DGCCA	0.228	0.343	0.377	0.357	0.202	+0.026

Table 7: **Verified differences between the paper and its reference implementation.** Each was confirmed in the source or empirically. The last two rows are defects rather than divergences.

Item	Location	Consequence
Singular entropy normalised by the L_2 norm, not L_1	<code>utilities/entropy.py</code>	Eq. 18 is not the entropy of a distribution. Shifts the selected diffusion time on 4/4 shipped datasets (MSRC 12 vs 1, Yale 8 vs 10, 100Leaves 5 vs 7, Caltech101-7 10 vs 17).
Kernel differs from the paper in two places	<code>benchmarks/utilities.py</code>	Paper: $\exp(-d/2\sigma^2)$. Code: $\exp(-d^2/bw)$. Squared distance, and bw in place of $2\sigma^2$.
k -NN count is $\lfloor \log N \rfloor$	<code>benchmarks/utilities.py</code>	Paper states $\lceil \log N \rceil$.
Q_X sign is internally inconsistent	paper Eq. 17 vs 19	Eq. 17 maximises Q ; Eq. 19 defines Q_X as a sum of negative log-probabilities. The code minimises it.
Q_{CH} scored on transition matrices, not data views	<code>mdt/mdt_direct.py</code>	Paper writes $CH(\mathbf{X}_v, C)$. Load-bearing: scoring on views instead drops MDT-DIRECT from 0.417 to 0.356.
MDT-Bsc grows the product on the opposite side	<code>mdt/_mdt_tree_utils.py</code>	<code>parent.path_operator @ op</code> yields $\mathbf{W}_1 \cdots \mathbf{W}_d$, reversing Def. 1. Same reachable set, different greedy prefix.
MDT-Cvx-RAND is not Dirichlet	<code>mdt/random_mdt.py</code>	Normalised i.i.d. uniforms concentrate near the simplex centre, i.e. nearer the uniform mean than a Dirichlet draw.
GCCA and MVSC implemented but unreported, and fed operators	<code>competitors/</code>	Both receive the $n \times n$ transition matrices as “views”. On raw views with a variance rank cut, GCCA rises from ≈ 0.03 to ≈ 0.68 AMI on MSRC-v5.
L-Isolet: two of its three views are identical	<code>benchmarks/isolet_lindenbaum.py</code>	K_1 and K_2 are written into one buffer and appended by reference, so the first is overwritten. Verified <code>allclose(view1, view2)</code> . The published L-Isolet has two distinct views, not three.
PCA and min-max fitted before the split	<code>benchmarks/mnist_*.py</code> , <code>olivetti.py</code>	Harmless transductively; leaks under any out-of-sample protocol.
Unseeded global RNG in <code>random_mdt</code>	<code>mdt/random_mdt.py</code>	Single MDT-RAND runs are not reproducible.